Bipedal Locomotion Control Based on Divergent Component of Motion(DCM)

Legged Robots Course

First Mini-project

Professor: Auke Ijspeert

Asimo runs at 9 Km/h by using the DCM concept for locomotion planning!

Takenaka, Toru, Takashi Matsumoto, and Takahide Yoshiike. "Real time motion generation and control for biped robot-1 st report: Walking gait pattern generation." 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, 2009.(Citation:299)

IHMC came in 2nd place in Darpa Robotics Challenge(DRC) by having the DCM planner that you will implement in the project!

Englsberger, Johannes, Christian Ott, and Alin Albu-Schäffer. "Three-dimensional bipedal walking control based on divergent component of motion." Ieee transactions on robotics 31.2 (2015): 355-368.(254)

Koolen, Twan, et al. "Design of a momentum-based control framework and application to the humanoid robot atlas." International Journal of Humanoid Robotics 13.01 (2016): 1650007.(160)

Linear Inverted Pendulum (LIP) model (2D

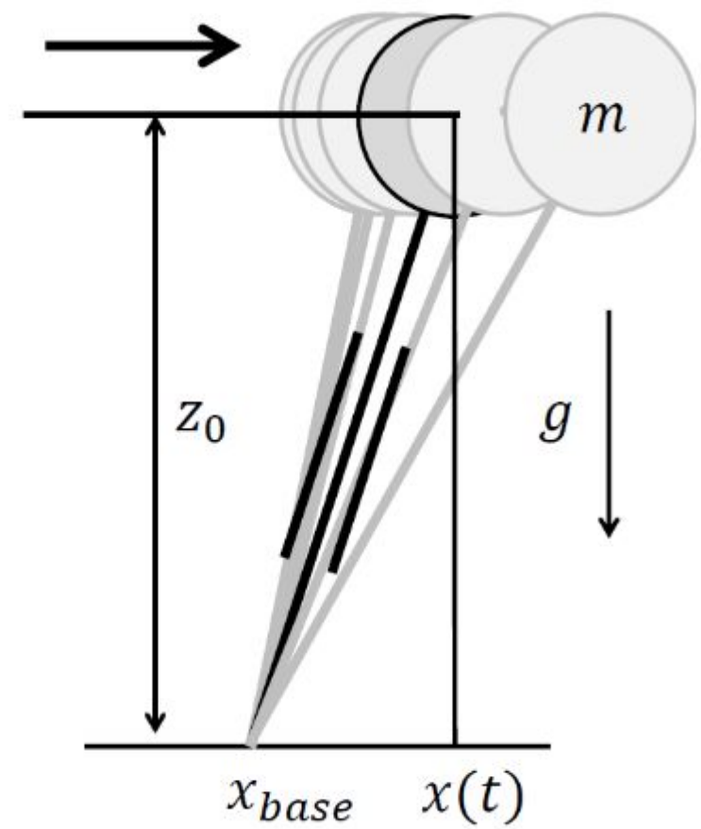
- Solving the equation of motion:

$$\ddot{x}(t) = \frac{g}{z_0}(x(t) - x_{base}),\ \omega = \sqrt{g/z_0}$$

$$x(t) = Ae^{-\omega t} + \boxed{Be^{\omega t}} + x_{base}$$

$$A = (-\dot{x}(0)/\omega + x(0) - x_{base})/2$$

$$B = \boxed{(\boxed{\dot{x}(0)/\omega + x(0)} - x_{base})/2}$$

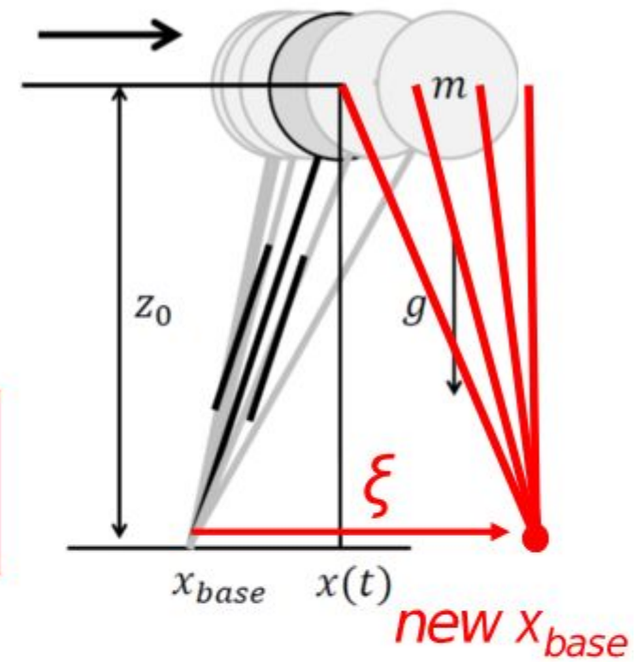


- We thus have a closed form (i.e. analytical solution) that allows us to predict the forward movement of the center of mass

Divergent component of motion (DCM)

- It is therefore important to monitor the DCM, and to plan the foot steps (i.e. x_{base}) to bring the robot (the CoM) where you want and to prevent falling.

$$\xi = x + \frac{\dot{x}}{\omega}$$



- The DCM corresponds to the ***instantaneous capture point.***
- Indeed by setting the next foot step (i.e. x_{base}) to ξ at the beginning of a step, the time evolution of x goes to zero (i.e. the pendulum stops).

$$\dot{\xi} = \omega(\xi - x_{base})$$

$$\dot{x} = -\omega(x - \xi)$$

This will be tested in the mini project with the Atlas simulation!

- Takenaka, Toru, Takashi Matsumoto, and Takahide Yoshiike. 2009. "Real Time Motion Generation and Control for Biped Robot: 1st Report: Walking Gait Pattern Generation." In *Proceedings of* IROS 2009.
- Englsberger et al (2014). Trajectory generation for continuous leg forces during double support and heel-to-toe shift based on divergent component of motion. In *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems* (pp. 4022-4029). IEEE.

Linear Inverted Pendulum Model(LIPM)-Divergent Component of Motion(DCM)

$$\ddot{x}_c = \omega^2(x_c - CoP)$$

x_c Center of mass position

CoP Center of Pressure

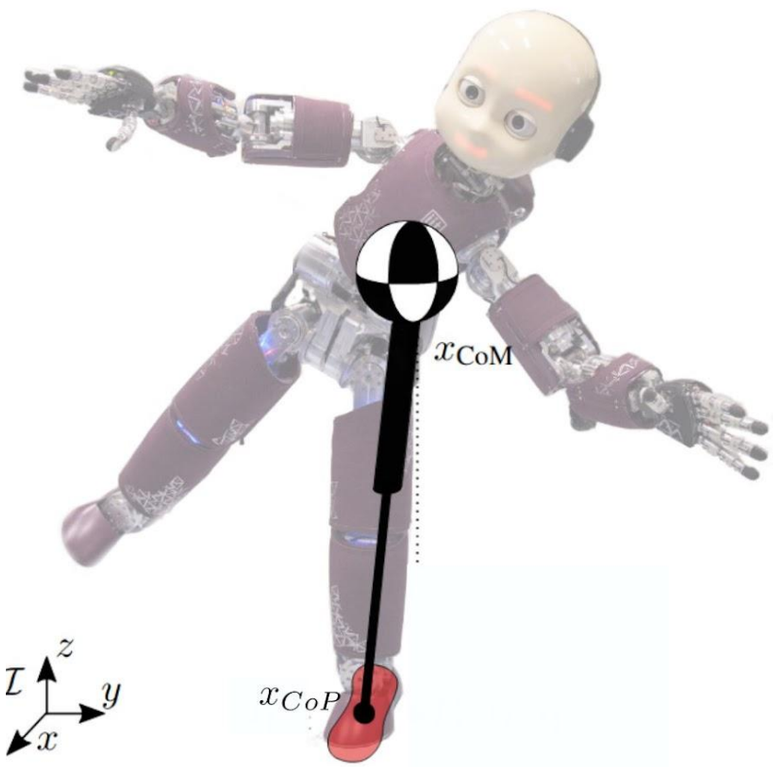
g Gravity Acceleration

z Center of mass height(constant)

$$\omega = \sqrt{\frac{g}{z}}$$

$$\xi_x = x_c + \frac{\dot{x}_c}{\omega}$$

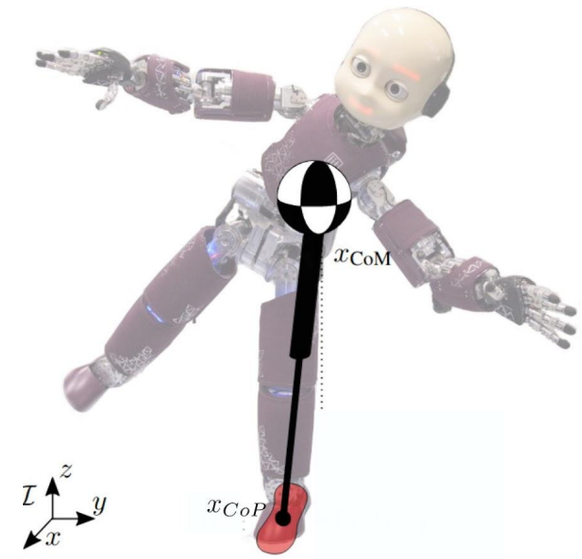
ξ DCM



Pratt, Jerry, et al. "Capture point:A step toward humanoid push recovery." 2006 6th IEEE-RAS international conference on humanoid robots. IEEE, 2006.(Citation:1013)

DCM Dynamics

$$\ddot{x}_c = \omega^2(x_c - CoP)$$



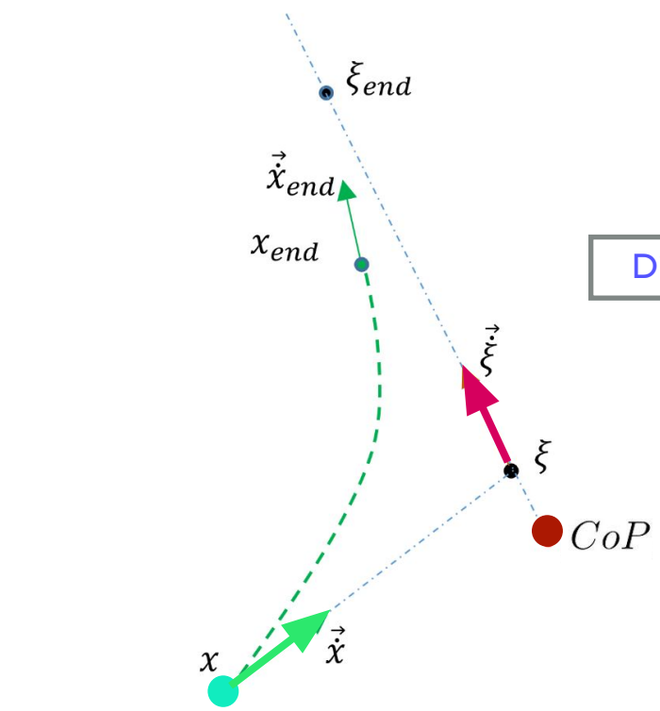
$$\xi_x = x_c + \frac{\dot{x}_c}{\omega}$$

$$\dot{\xi} = \omega(\xi - CoP)$$

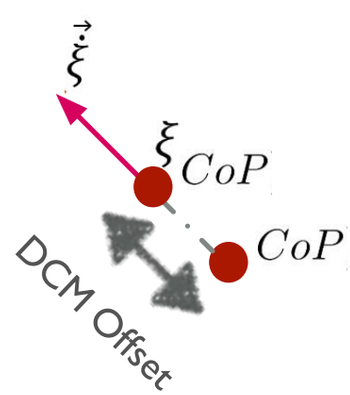
Divergent Component of Motion(DCM)

$$\dot{x} = -\omega(x - \xi)$$

Convergent Component of Motion



Why is capture point another name for DCM?



Koolen, Twan, et al. "Capturability-based analysis and control of legged locomotion, Part 1: Theory and application to three simple gait models." The international journal of robotics research 31.9 (2012): 1094-1113.(Citation:433)

Solution of DCM ode

$$\dot{\boldsymbol{\xi}} = \omega(\boldsymbol{\xi} - \mathbf{cop})$$

$$\downarrow$$

$$\dot{\boldsymbol{\xi}} - \omega\boldsymbol{\xi} = -\omega\,\mathbf{cop}_0$$

$$\downarrow$$

We can solve this ode by considering that CoP is a constant value (During single support phase)

$$\downarrow$$

$$\boldsymbol{\xi}(t) = e^{\int \omega dt}\left[\int (-\mathbf{cop}_0\,\omega) e^{\int -\omega dt} dt + \mathbf{C}\right],$$

Solution of DCM ode

$$\boldsymbol{\xi}(t) = e^{\int \omega dt}\left[\int (-\mathbf{cop}_0\,\omega) e^{\int -\omega dt} dt + \mathbf{C}\right],$$

$$\boldsymbol{\xi}(0) = \boldsymbol{\xi}_0 = \mathbf{cop}_0 + \mathbf{C}_0, \qquad \boldsymbol{\xi}(T) = \boldsymbol{\xi}_T = \mathbf{cop_0} + \mathbf{C_f}\, e^{\omega T}.$$

$$\boldsymbol{\xi}_0 - \mathbf{cop}_0 = (\boldsymbol{\xi}_T - \mathbf{cop}_0) e^{-\omega T}.$$

$$\sigma = e^{\omega T} \qquad \boldsymbol{\gamma_T} = \boldsymbol{\xi_T} - \mathbf{cop_T}$$

$$\boldsymbol{\gamma_T} + \mathbf{cop}_T + (\mathbf{cop}_0 - \boldsymbol{\xi}_0)\sigma = \mathbf{cop}_0$$

Optimization based on QP

$$J = \alpha_1 \left\| \mathbf{cop}_T - \mathbf{cop}_{T,nom} \right\|^2 + \alpha_2 \| \boldsymbol{\gamma}_T - \boldsymbol{\gamma}_{nom} \|^2 + \alpha_3 | \sigma - e^{\omega T_{nom}} |^2,$$

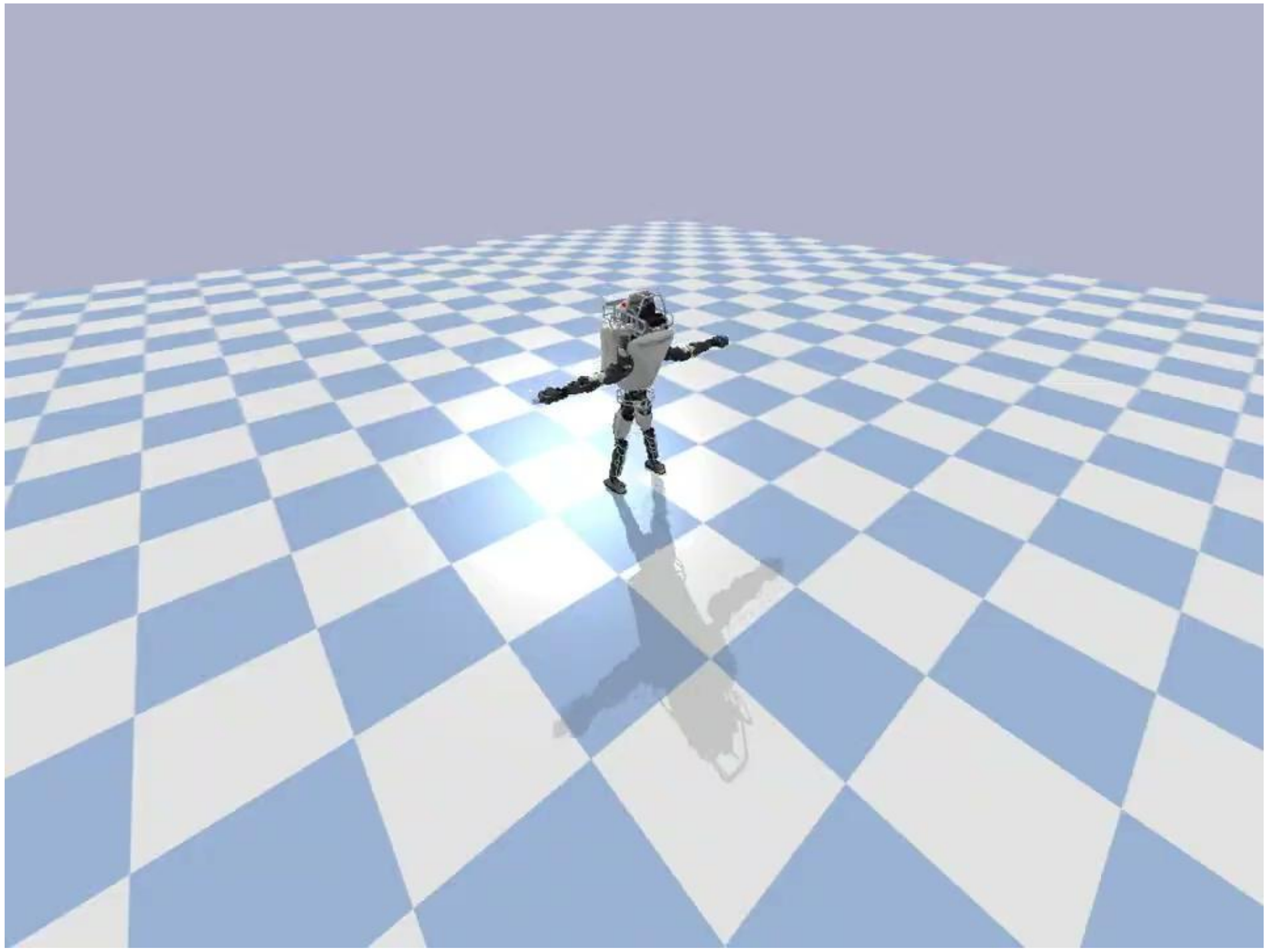
$$\begin{bmatrix} I_2 & 0_{2\times1} & 0_2 \\ -I_2 & 0_{2\times1} & 0_2 \\ 0_{1\times2} & I_1 & 0_{1\times2} \\ 0_{1\times2} & -I_1 & 0_{1\times2} \end{bmatrix} \begin{bmatrix} \mathbf{cop}_T \\ \sigma \\ \boldsymbol{\gamma}_T \end{bmatrix} \leq \begin{bmatrix} \mathbf{cop}_{T,max} \\ -\mathbf{cop}_{T,min} \\ \sigma_{max} \\ -\sigma_{min} \end{bmatrix},$$

$$\boxed{\boldsymbol{\gamma}_T} + \boxed{\mathbf{cop}_T} + (\mathbf{cop}_0 - \boldsymbol{\xi}_0)\boxed{\sigma} = \mathbf{cop}_0$$

Quadratic Programming

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \frac{1}{2} x^T P x + q^T x \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array}$$

Optimization based on QP



- Please clone this repo: https://gitlab.epfl.ch/shafiee/lr-dcm-start.git
- This assignment: 15% of course grade.
- Upload your code and report.